Department of Mechanical and Production Engineering

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Exercises

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Lecture 1: Fourier Theory

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1.1 Exercise 1.

Consider the following periodic functions. Find possible values for the period, using the definition of periodic functions.

(a) $f(x) = e^{\cos x}$

(b) $f(x) = \cos\left(x + \frac{\pi}{3}\right)$

(c) $f(x) = \cos\frac{\pi x}{3}$

(d) $f(x) = \cos x + \cos 3x$

(e) $f(x) = \cos x \cos 3x$

(f) $f(x) = \cos x + \cos(0{,}6x)$

For (a) the exponent to e, i.e. $\cos x$ only has a range of $(-1;1)$. This means that the exponent will change between -1 and 1 with a period of 2π, therefore the period $p_a = 2\pi$ will also be the case for the function described in (a).

For (b) we are once again working with a cos function. This time it is however shifted by $\frac{\pi}{3}$ along the x-axis. A translation along the x-axis will, however, not affect the period of the function and therefore the period here is once again $p_b = 2\pi$.

For (c) the x is multiplied by $\frac{\pi}{3} \approx 1{,}05$. This means that the argument to the cosine will reach 2π at a speed that is $\frac{\pi}{3} \approx 1{,}05$ times faster than a normal cosine. Therefore the period of this is $p_c = \frac{3\cdot 2\pi}{\pi} = 6$

For (d) we use that

$$f(x) = f(x+p)$$

for a periodic function to get

$$\begin{aligned}\cos x + \cos 3x &= \cos\left(x+p\right) + \cos\left(3\left(x+p\right)\right)\\ \cos x + \cos 3x &= \cos(x+p) + \cos\left(3x+3p\right).\end{aligned}$$

We see that this is true when both p and $3p$ are integer multiples of 2π so $p_d = 2\pi$

For (e) we use the same definition to get

$$\cos x \cos 3x = \cos\left(x+p\right)\cos\left(3x+3p\right).$$

We see that this is also true only when both p and $3p$ are integer multiples of 2π so $p_e = 2\pi$

For (f) we can once again use the same definition to get

$$\cos x + \cos(0{,}6x) = \cos\left(x+p\right) + \cos\left(0{,}6x + 0{,}6p\right).$$

This is true when p and $0{,}6p$ are integer multiples of 2π, so $p_f = 10\pi$.

1.2 Exercise 2.

Let

$$f(x) = x, \quad -1 \le x \le 1, \quad f(x) = f(x+p), \quad p = 2.$$

- Sketch f.
- Find a Fourier series representation of f.

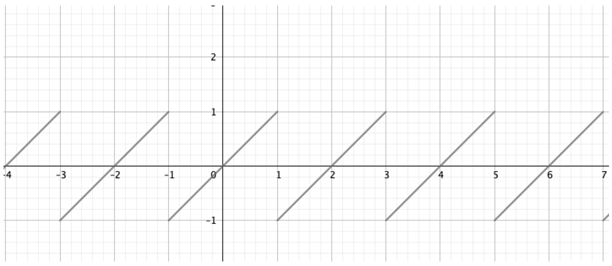

Figure 0.1

f has been sketched on **Figure 0.1**. The Fourier series of this can, as it is a “nice” function be found by the formula:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right).$$

Where $L = \frac{p}{2} = 1$ with $p = 2$ being the period of the function. The Fourier coefficients can be found by the Euler formulas:

$$\begin{aligned}
a_0 &= \frac{1}{2L} \int_{-L}^{L} f(x)\,\mathrm{d}x \\
a_n &= \frac{1}{L} \int_{-L}^{L} f(x) \cos \frac{n\pi x}{L}\,\mathrm{d}x, \quad n = 1, 2, 3, \ldots \\
b_n &= \frac{1}{L} \int_{-L}^{L} f(x) \sin \frac{n\pi x}{L}\,\mathrm{d}x, \quad n = 1, 2, 3, \ldots.
\end{aligned}$$

We start by solving for a_0 as:

$$a_0 = \int_{-1}^{1} x\,\mathrm{d}x = \left[\frac{1}{2} x^2 \right]_{-1}^{1} = \cdot 0 = 0.$$

We now solve for a_n using integration by parts as:

$$\begin{aligned}
a_n &= \int_{-1}^{1} x \cos n\pi x\,\mathrm{d}x \\
&= \left[x \frac{\sin n\pi x}{n\pi} \right]_{-1}^{1} - \int_{-1}^{1} \frac{\sin n\pi x}{n\pi}\,\mathrm{d}x \\
&= \left(-\frac{1}{n\pi} \right) \int_{-1}^{1} \sin n\pi x\,\mathrm{d}x \\
&= \left[-\frac{1}{n\pi} \right] \left[-\frac{\cos n\pi x}{n\pi} \right]_{-1}^{1} \\
&= 0.
\end{aligned}$$

We similarly solve for b_n as:

$$\begin{aligned}
b_n &= \int_{-1}^{1} x \sin n\pi x\,\mathrm{d}x \\
&= \left[x \frac{-\cos n\pi x}{n\pi} \right]_{-1}^{1} - \int_{-1}^{1} \frac{-\cos n\pi x}{n\pi}\,\mathrm{d}x \\
&= \left(-\frac{2}{n\pi} \right) \cos n\pi + \frac{1}{n\pi} \left[\frac{\sin n\pi x}{n\pi} \right]_{-1}^{1} \\
&= \left(-\frac{2}{n\pi} \right) \cos n\pi.
\end{aligned}$$

The full Fourier series therefore becomes:

$$f(x) = \sum_{n=1}^{\infty} \left(-\frac{2}{n\pi} \right) \cos n\pi \cdot \sin n\pi x.$$

1.3 Exercise 3.

Let $a \neq 0$ and $b \neq 0$. Let $f(x) = f(x + p_0)$ be a periodic function with period p_0. Is $f(ax + b)$ periodic? If yes, find a period. Give arguments for your answer.

The b only serves to translate the function $f(x)$ along the x-axis, this does not change the period. The value of a corresponds to how "quickly" some value is reached, e.g. if ax is an argument to a function a doubling of the value of a will mean that the function will reach its maximum for a value of x only half as big. I.e. there is an inverse relation between the size of a and the period as:

$$p_2 = \frac{p_1}{a}.$$

1.4 Exercise 4.

Calculate the left hand limit and the right hand limit of

$$f(x) = \frac{|x|}{x}$$

at $x = 0$.

We start with the right-hand limit as:

$$f(0+) = \lim_{h\to 0}(0 + h) = \lim_{h\to 0}\frac{|h|}{h} = \lim_{h\to 0}\frac{h}{h} = 1.$$

And now we can similarly do the left-hand limit as:

$$f(0-) = \lim_{h\to 0}(0 - h) = \lim_{h\to 0}\frac{|-h|}{-h} = \lim_{h\to 0}\frac{h}{-h} = -1.$$

1.5 Exercise 5.

Calculate the left-hand derivative and the right-hand derivative of

$$f(x) = x|x|$$

at $x = 0$.

We start with the right-hand derivative as:

$$f'(0+) = \lim_{h\to 0}\frac{f(0+h) - f(0)}{h} = \lim_{h\to 0}\frac{h|h|}{h} = \lim_{h\to 0} h = 0.$$

And the left-hand derivative can likewise be computed as:

$$f'(0-) = \lim_{h\to 0}\frac{f(0) - f(0-h)}{h} = \lim_{h\to 0}\frac{-(-h)|-h|}{h} = \lim_{h\to 0} h = 0.$$

Lecture 2: Even and odd functions

5. September 2025

2.1 Exercise 1.

Find a period and the corresponding Fourier coefficients of

a) $f(x) = \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{2x}{4}\right) + \cos\left(\frac{\pi}{5}\right)\sin\left(\frac{3x}{7}\right)$

We rewrite the given function as:

$$
\begin{aligned}
f(x) &= \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{2x}{4}\right) + \cos\left(\frac{\pi}{5}\right)\sin\left(\frac{3x}{7}\right) \\
&= \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{7\pi x}{14\pi}\right) + \cos\left(\frac{\pi}{5}\right)\sin\left(\frac{6\pi x}{14\pi}\right).
\end{aligned}
$$

We will now compare this to the general formulation of the Fourier series as:

$$
f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right).
$$

We can quickly see that $L = 14\pi \implies p = 28\pi$, $b_6 = \cos\left(\frac{\pi}{5}\right)$, $b_7 = \sin\left(\frac{\pi}{4}\right)$. All other unmentioned Fourier coefficients are zero.

b) $f(x) = \sin\left(\sqrt{2}x\right) + \cos\left(\frac{x}{\sqrt{2}}\right)$

We once again start by rewriting as:

$$
\begin{aligned}
f(x) &= \sin\left(\sqrt{2}x\right) + \cos\left(\frac{x}{\sqrt{2}}\right) \\
&= \sin\left(\frac{2\pi x}{\sqrt{2}\pi}\right) + \cos\left(\frac{\pi x}{\sqrt{2}\pi}\right).
\end{aligned}
$$

Comparing this with the general formulation of the Fourier series:

$$
f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right).
$$

We see that $L = \sqrt{2}\pi \implies p = 2\sqrt{2}\pi$, $a_1 = 1$, $b_2 = 1$. All other Fourier coefficients are zero.

c) $f(x) = \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right)$

We use that

$$
\sin(\alpha+\beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha
$$

to rewrite the equation as:

$$
\begin{aligned}
f(x) &= \sin\left(\frac{\pi}{3} + \frac{3\pi x}{4}\right) \\
&= \sin\frac{\pi}{3}\cdot\cos\frac{3\pi x}{4} + \sin\frac{3\pi x}{4}\cdot\cos\frac{\pi}{3}.
\end{aligned}
$$

Comparing this to the general formulation of the Fourier series

$$
f(x) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right)\right)
$$

we see that $L = 4 \implies p = 8$, $a_3 = \sin\frac{\pi}{3}$, $b_3 = \cos\frac{\pi}{3}$. All other Fourier coefficients are zero

2.2 Exercise 2.

Are the following functions even or odd? Give arguments.

a) $f(x) = \left(\cos x^2 + \cos x\right) \sin x$

We check if the function is even or odd by checking what happens for an input of $-x$ as:

$$\begin{aligned} f(-x) &= \left(\cos(-x)^2 + \cos(-x)\right) \sin(-x) \\ &= \left(\cos x^2 + \cos(-x)\right)(-\sin x) \\ &= -f(x). \end{aligned}$$

Hence the function is odd.

b) $f(x) = \frac{1+\cos x}{2+\cos x^2}$

We follow the same procedure:

$$\begin{aligned} f(-x) &= \frac{1+\cos(-x)}{2+\cos(-x)^2} \\ &= \frac{1+\cos x}{2+\cos x} \\ &= f(x). \end{aligned}$$

Hence the function is even.

c) $f(x) = \frac{1+\sin x}{2+\sin x^2}$

The same procedure is applied:

$$\begin{aligned} f(-x) &= \frac{1+\sin(-x)}{2+\sin(-x)^2} \\ &= \frac{1-\sin x}{2+\sin x} \\ &\neq \begin{cases} f(x) \\ -f(x) \end{cases} \end{aligned}$$

Hence the function is neither even nor odd.

d) $f(x) = x|x|$

The same procedure is applied:

$$\begin{aligned} f(-x) &= (-x)\,|-x| \\ &= -x\,|x| \\ &= -f(x). \end{aligned}$$

Hence the function is odd.

2.3 Exercise 3.

Consider the periodic function

$$f(x) = \begin{cases} x, & -2 < x < -1 \\ x + k, & -1 < x < 1 \\ x, & 1 < x < 2 \end{cases}$$
$$f(x) = f(x+p), p = 4$$

where k is a constant. Decompose $f(x)$ into its even and odd part.

We apply the definition of decomposition into odd and even parts for the first interval, $-2 < x < -1$ as:

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}(x + (-x)) = 0$$
$$f_2(x) = \frac{1}{2}(f(x) - f(-x) = \frac{1}{2}(x - (-x)) = x.$$

For the second interval, $-1 < x < 1$ we get

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}((x + k) + (-x + k)) = k$$
$$f_2(x) = \frac{1}{2}(f(x) - f(-x) = \frac{1}{2}((x + k) - (-x + k)) = x.$$

And for the last interval, $1 < x < 2$ we get:

$$f_1(x) = \frac{1}{2}(f(x) + f(-x)) = \frac{1}{2}(x + (-x)) = 0$$
$$f_2(x) = \frac{1}{2}(f(x) - f(-x) = \frac{1}{2}(x - (-x)) = x.$$

We thus have:

$$f(x) = f_1(x) + f_2(x)$$
$$f_1(x) = \begin{cases} 0 & -2 < x < -1 \\ k & -1 < x < 1 \\ 0 & 1 < x < 2. \end{cases}$$
$$f_1(x) = f_1(x+p), \quad p = 4$$
$$f_2(x) = x, \qquad -2 < x < 2$$
$$f_2(x) = f_2(x+p), \quad p = 4.$$

2.4 Exercise 4.

Consider the periodic function

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 2x, & 0 < x < \pi \end{cases}$$
$$f(x) = f(x+p), p = 2\pi.$$

Decompose $f(x)$ into its even and odd part and find its Fourier series.

Hint: You may want to use examples from the lecture.

For the interval $-\pi < x < 0$ we have

$$f_1(x) = \frac{1}{2}(0 + (-2x)) = -x$$
$$f_2(x) = \frac{1}{2}(0 - (-2x)) = x.$$

And for $0 < x < \pi$ we have

$$\begin{aligned} f_1(x) &= \frac{1}{2}(2x + 0) = x \\ f_2(x) &= \frac{1}{2}(2x - 0) = x. \end{aligned}$$

We thus have

$$\begin{aligned} &f(x) = f_1(x) + f_2(x) \\ &f_1(x) = |x|, -\pi < x < \pi \\ &f_1(x) = f_1(x+p), \quad p = 2\pi \\ &f_2(x) = x, \quad -\pi < x < \pi \\ &f_2(x) = f_2(x+p) \quad p = 2\pi. \end{aligned}$$

From the lecture we know that f_1 has the Fourier cosine series

$$f_1(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2}(\cos n\pi - 1)\cos nx$$

and that f_2 has the Fourier sine series:

$$f_2(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{2}{n} \sin nx.$$

Combining these we get the Fourier series for $f(x)$ as:

$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \left(\frac{2}{\pi n^2}(\cos n\pi - 1)\cos nx + (-1)^{n+1}\frac{2}{n}\sin nx \right).$$

2.5 Exercise 5.

Decompose

$$f(x) = e^{ix}$$

into its even and odd part.

Hint; you may want to use the representation of $f(x)$ in terms of $\cos x$ and $\sin x$.

We apply the same procedure as in the last couple of exercises whilst remembering that:

$$e^{ix} = \cos x + i \sin x.$$

We therefore get:

$$\begin{aligned} f_1(x) &= \frac{1}{2}(f(x)+f(-x)) \\ &= \frac{1}{2}(\cos x + i\sin x + \cos x - i\sin x) \\ &= \cos x \\ f_2(x) &= \frac{1}{2}\left(f(x)-f(-x)\right) \\ &= \frac{1}{2}(\cos x + i\sin x - (\cos x - i\sin x)) \\ &= i\sin x. \end{aligned}$$

This also is what would be expected when looking at the representation of e^{ix} introduced at the start of this answer.

Lecture 3: Fourier Series

12. September 2025

3.1 Exercise 1.

Consider the function $f(x) = \cos x, 0 < x < \pi$.

Hint: In some cases the expansion can be an even/odd function.

Hint: For $m, n \in \mathbb{N}, m \neq n, L > 0$:

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \cos x \sin(2nx)\,\mathrm{d}x &= -\frac{2n}{1-4n^2} \\ \int_0^L \cos\left(\frac{m\pi x}{L}\right)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x &= \frac{nL}{\pi\left(m^2-n^2\right)}(\cos(n\pi)\cos(m\pi)-1) \\ \int_0^L \cos\left(\frac{\pi x}{L}\right)\sin\left(\frac{\pi x}{L}\right)\mathrm{d}x &= 0. \end{aligned}$$

(a) Sketch the expansion of f and find the corresponding Fourier series.

The expansion of f is simply the "repetition" of f indefinitely along the x-axis. This is shown on Figure 0.2

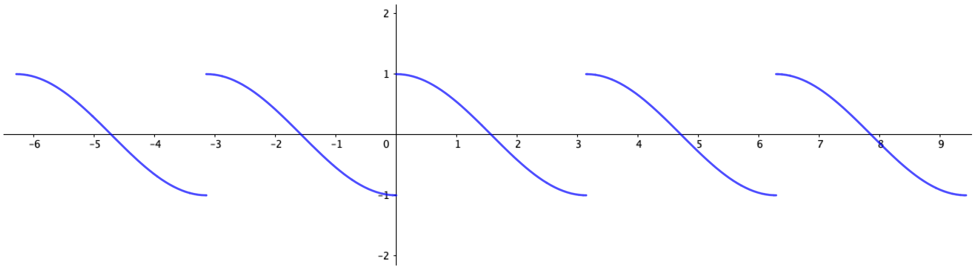

Figure 0.2: Expansion of $f(x) = \cos x, 0 < x < \pi$

To find the Fourier series of this we must first define a periodic function $f_0(x)$ around $x = 0$ based on Figure 0.2 we see that this can be written as

$$f_0(x) = \begin{cases} \cos x, & 0 < x < \frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2} < x < 0. \end{cases}$$

$$f_0(x) = f_0(x+p), p = 2L = \pi.$$

As $f_0(x)$ is an odd function its Fourier series will be a so-called Fourier sine series:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right).$$

Where the Fourier coefficient b_n is

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x.$$

Substituting the general $f(x)$ for our $f_0(x)$ we get:

$$\begin{aligned} b_n &= \frac{4}{\pi}\int_0^{\frac{\pi}{2}} \cos x \sin(2nx)\,\mathrm{d}x \\ &= \frac{4}{\pi}\cdot\left(-\frac{2n}{1-4n^2}\right) \\ &= \frac{8n}{\pi\left(4n^2-1\right)}. \end{aligned}$$

Hence the Fourier series becomes:

$$f_0(x) = \sum_{n=1}^{\infty} \frac{8n}{\pi\left(4n^2-1\right)}\sin(2nx).$$

(b) Sketch the even expansion of f and find the corresponding Fourier cosine series.

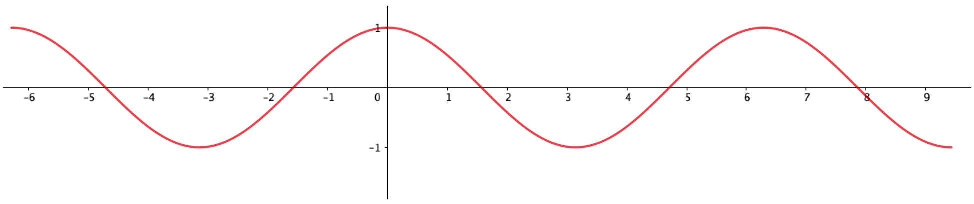

Figure 0.3: Even expansion of $f(x) = \cos x, 0 < x < \pi$

On Figure 0.3 it can be seen that the even expansion of the expression is simply the cos function. We can therefore write our periodic function as

$$f_1(x) = \cos x, \quad 0 < x < \frac{\pi}{2}$$
$$f_1(x) = f_1(x+p), \quad p = 2L = 2\pi.$$

This can also be written as

$$f_1(x) = \cos x.$$

We remember that the general form of a Fourier cosine series is

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right)$$

or with the values given in the exercise

$$f_1(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{\pi}\right) = a_0 + \sum_{n=1}^{\infty} a_n \cos nx.$$

Now we can compare these

$$\cos x = a_0 + \sum_{n=1}^{\infty} a_n \cos nx$$
$$\implies a_0 = 0, a_1 = 1.$$

(c) Sketch the odd expansion of f and find the corresponding Fourier sine series. The odd expansion gives the

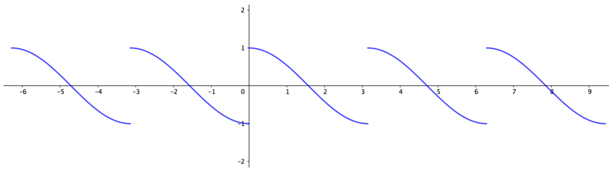

Figure 0.4: Odd expansion of $f(x) = \cos x, 0 < x < \pi$

periodic function

$$f_2(x) = \begin{cases} \cos x, & 0 < x < \pi \\ -\cos x & -\pi < x < 0. \end{cases}$$
$$f_2(x) = f_2(x+p), \quad p = 2L = 2\pi.$$

We remember that the general form of a Fourier sine series is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

or with the values in the exercise

$$f_2(x) = \sum_{n=1}^{\infty} b_n \sin(nx)$$

where b_n is

$$b_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right) \mathrm{d}x$$

or

$$b_n = \frac{2}{\pi}\int_0^{\pi} \cos x \sin(nx)\, \mathrm{d}x.$$

For $n = 1$ this simplifies to

$$b_1 = 0.$$

For $n > 1$ it becomes

$$b_n = \frac{2}{\pi} \cdot \frac{n\pi}{\pi\left(1-n^2\right)}(-\cos(n\pi) - 1)$$
$$= \frac{2n}{\pi\left(n^2-1\right)}(1 - \cos(n\pi)).$$

Hence the full Fourier sine series is

$$f_2(x) = \sum_{n=2}^{\infty} \frac{2n}{\pi\left(n^2-1\right)}(1 + \cos(n\pi))\sin(nx).$$

(d) Calculate the error E_N of the expansion and the odd expansion for $N = 1, 2, 3$. What can be said about the quality of the approximations for these two expansions?

The general formula for calculating the error E_N of the approximation is

$$E_N = \int_{-L}^{L} f^2(x)\,\mathrm{d}x - L\left(2a_0^2 + \sum_{n=1}^{N}\left(a_n^2 + b_n^2\right)\right).$$

If we start by considering the *expansion* we are working with the expression

$$f_0(x) = \sum_{n=1}^{\infty} \frac{8n}{\pi\left(4n^2-1\right)} \sin(2nx).$$

Over the entire interval $[-L;L]$ $f_0^2 = \cos^2$. Using this fact and substituting $L = \frac{\pi}{2}$, $a_n = 0$ and $b_n = \frac{8n}{\pi(4n^2-1)}$ we can write

$$\begin{aligned}
E_N &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2(x)\,\mathrm{d}x - \frac{\pi}{2}\sum_{n=1}^{N}\left(\frac{8n}{\pi\left(4n^2-1\right)}\right)^2 \\
&= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2}\left(1+\cos(2x)\right)\mathrm{d}x - \frac{\pi}{2}\sum_{n=1}^{N}\frac{64n^2}{\pi^2\left(4n^2-1\right)^2} \\
&= \frac{\pi}{2} + \frac{1}{2}\left[\frac{1}{2}\sin(2x)\right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \frac{32}{\pi}\sum_{n=1}^{N}\frac{n^2}{\left(4n^2-1\right)^2} \\
&= \frac{\pi}{2} - \frac{32}{\pi}\sum_{n=1}^{N}\frac{n^2}{\left(4n^2-1\right)^2}.
\end{aligned}$$

Calculating the numerical value of this for $n = 1,2,3$ yields

$$\begin{aligned}
E_1 &= \frac{\pi}{2} - \frac{32}{9\pi} \approx 0{,}439 \\
E_2 &= \frac{\pi}{2} - \frac{32}{\pi}\left(\frac{1}{9} + \frac{4}{15^2}\right) \approx 0{,}258 \\
E_3 &= \frac{\pi}{2} - \frac{32}{\pi}\left(\frac{1}{9} + \frac{4}{15^2} + \frac{9}{35^2}\right) \approx 0{,}183.
\end{aligned}$$

Now moving on to the odd expansion our $f_2^2(x) = \cos^2(x)$ as before, however this time $L = \pi$ and $b_n = \frac{2n}{\pi(n^2-1)}\left(1+\cos(n\pi)\right)$. Plugging these values into the formula we get:

$$\begin{aligned}
E_N &= \int_{-\pi}^{\pi} \cos^2(x)\,\mathrm{d}x - \pi\sum_{n=2}^{N}\left(\frac{2n}{\pi\left(n^2-1\right)}\left(1+\cos(n\pi)\right)\right)^2 \\
&= \pi - \pi\sum_{n=2}^{N}\frac{4n^2}{\pi^2\left(n^2-1\right)^2}\left(1+\cos(n\pi)\right)^2 \\
&= \pi - \frac{4}{\pi}\sum_{n=2}^{N}\frac{n^2}{\left(n^2-1\right)^2}\left(1+\cos(n\pi)\right)^2.
\end{aligned}$$

And for $n = 1,2,3$ we get

$$\begin{aligned}
E_1 &= \pi - \frac{4}{\pi}\cdot 0 = \pi \\
E_2 &= \pi - \frac{4}{\pi}\left(0 + \frac{16}{9}\right) \approx 0{,}878 \\
E_3 &= \pi - \frac{4}{\pi}\left(0 + \frac{16}{9} + \frac{9}{64}\cdot 0\right) \approx 0{,}878.
\end{aligned}$$

When comparing errors we must take into account the length of the intervals L. Therefore a realistic comparison requires us to half the errors from the odd expansion. Even when doing so the error rate from the “normal” expansion performs better and should therefore be preferred.

3.2 Exercise 2.

Find the Fourier integral representation of

$$f(x) = \begin{cases} 1, & -1 < x < 1 \\ -1, & 1 < x < 2 \\ -1, & -2 < x < -1 \\ 0, & \text{otherwise.} \end{cases}.$$

The Fourier integral is given by

$$f(x) = \int_0^\infty (A(\omega)\cos(\omega x) + B(\omega)\sin(\omega x))\, \mathrm{d}\omega$$

where

$$A(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos(\omega x)\, \mathrm{d}x$$

and

$$B(\omega) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\sin(\omega x)\, \mathrm{d}x.$$

We start by finding $A(\omega)$ using piecewise integration

$$\begin{aligned} A(\omega) &= \frac{1}{\pi}\left(\int_{-2}^{-1} -\cos(\omega x)\, \mathrm{d}x + \int_{-1}^{1} \cos(\omega x)\, \mathrm{d}x + \int_{1}^{2} -\cos(\omega x)\, \mathrm{d}x\right) \\ &= \frac{4\sin\omega - 2\sin(2\omega)}{\pi\omega}. \end{aligned}$$

The same technique can now be applied to find $B(\omega)$ as

$$\begin{aligned} B(\omega) &= \frac{1}{\pi}\left(\int_{-2}^{-1} -\sin(\omega x)\, \mathrm{d}x + \int_{-1}^{1} \sin(\omega x)\, \mathrm{d}x + \int_{1}^{2} -\sin(\omega x)\, \mathrm{d}x\right) \\ &= 0. \end{aligned}$$

Therefore the Fourier integral representation of $f(x)$ is

$$f(x) = \int_0^\infty \left(\frac{4\sin\omega - 2\sin(2\omega)}{\pi\omega}\right)\cos(\omega x)\, \mathrm{d}\omega.$$

3.3 Exercise 3.

Find the Fourier integral representation of

$$f(x) = \begin{cases} -1, & -2 < x < 0 \\ 1, & 0 < x < 2 \\ 0, & \text{otherwise.} \end{cases}.$$

Here the same procedure can be applied as above to get

$$\begin{aligned} A(\omega) &= \frac{1}{\pi}\left(\int_{-2}^{0} -\cos(\omega x)\, \mathrm{d}x + \int_{0}^{2} \cos(\omega x)\, \mathrm{d}x\right) \\ &= 0. \end{aligned}$$

and

$$B(\omega) = \frac{1}{\pi}\left(\int_{-2}^{0} -\sin(\omega x)\,\mathrm{d}x + \int_{0}^{2} \sin(\omega x)\,\mathrm{d}x\right)$$
$$= \frac{2-2\cos(2\omega)}{\pi\omega}.$$

Which becomes

$$f(x) = \int_0^\infty \frac{2-2\cos(2\omega)}{\pi\omega}\sin(\omega x)\,\mathrm{d}\omega.$$

3.4 Exercise 4.

Consider the periodic rectangular wave and its Fourier series as discussed in Lecture 1. Calculate the error of the N-th approximation for $N = 1,2,3$.

The general formula for calculating the error E_N of the approximation is

$$E_N = \int_{-L}^{L} f^2(x)\,\mathrm{d}x - L\left(2a_0^2 + \sum_{n=1}^{N}\left(a_n^2 + b_n^2\right)\right).$$

In this case $f(x)^2 = k^2$, $L = \pi$, $a_0 = 0$, $a_n = 0$, and $b_n = \frac{2k}{n\pi}(1-\cos(n\pi))$. I.e.

$$E_N = \int_{-\pi}^{\pi} k^2\,\mathrm{d}x - \pi\sum_{n=1}^{\infty}\left(\frac{2k}{n\pi}(1-\cos(n\pi))\right)^2$$
$$= 2\pi k^2 - \frac{4k^2}{\pi}\sum_{n=1}^{N}\frac{1}{n^2}(1-\cos(n\pi))^2.$$

For $n = 1,2,3$ we get $E_1 = E_2 \approx 1{,}19k^2$ and $E_3 \approx 0{,}62k^2$.

Lecture 4: Fourier sine and cosine transformations

19. September 2025

4.1 Exercise 1.

Represent $f(x)$, $0 < x < \infty$, defined as

$$f(x) = \begin{cases} \cos x, & 0 < x < \pi \\ 0, & x > \pi. \end{cases}$$

as a Fourier sine integral. *Hint:*

$$\int_0^\pi \cos x \sin(\omega x)\,\mathrm{d}x = \frac{\omega(1+\cos(\omega\pi))}{\omega^2-1}$$

We start by determining the value of $B(\omega)$ as

$$B(\omega) = \frac{2}{\pi}\int_0^\pi f(x)\sin(\omega x)\,\mathrm{d}x$$
$$= \frac{2}{\pi}\int_0^\pi \cos x\sin(\omega x)\,\mathrm{d}x$$
$$= \frac{2}{\pi}\cdot\frac{\omega(1+\cos(\omega\pi))}{\omega^2-1}$$

.

Hence, the Fourier sine integral becomes

$$f(x) = \int_0^\infty B(\omega)\sin(\omega x)\,\mathrm{d}\omega = \int_0^\infty \frac{2}{\pi}\frac{\omega(1+\cos(\omega\pi))}{\omega^2-1}\sin(\omega x)\,\mathrm{d}\omega.$$

4.2 Exercise 2.

Represent $f(x)$, $0 < x < \infty$, defined as

$$f(x) = \begin{cases} \sin x, & 0 < x < \pi \\ 0, & x > \pi \end{cases}$$

as a Fourier cosine integral. *Hint:*

$$\int_0^\pi \sin x \cos(\omega x)\,\mathrm{d}x = \frac{1+\cos(\omega\pi)}{1-\omega^2}.$$

We determine the value of $A(\omega)$ as

$$\begin{aligned} A(\omega) &= \frac{2}{\pi}\int_0^\infty f(x)\cos(\omega x)\,\mathrm{d}x \\ &= \frac{2}{\pi}\int_0^\pi \sin x\cos(\omega x) \\ &= \frac{2}{\pi}\frac{1+\cos(\omega\pi)}{1-\omega^2}. \end{aligned}$$

Hence, the Fourier cosine integral becomes

$$f(x) = \int_0^\infty A(\omega)\sin(\omega x)\,\mathrm{d}\omega = \int_0^\infty \frac{2}{\pi}\frac{1+\cos(\omega\pi)}{1-\omega^2}\cos(\omega x)\,\mathrm{d}\omega.$$

4.3 Exercise 3.

Find the Fourier cosine transform of $f(x), 0 < x < \infty$, defined as

$$f(x) = \begin{cases} 1, & 0 < x < 1 \\ -1 & 1 < x < 2 \\ 0, & x > 2 \end{cases}.$$

We have the general formula for the cosine transform

$$\begin{aligned} \mathscr{F}_C(f) &= \sqrt{\frac{2}{\pi}}A(\omega) \\ &= \sqrt{\frac{2}{\pi}}\left(\int_0^1 \cos(\omega x)\,\mathrm{d}x - \int_1^2 \cos(\omega x)\,\mathrm{d}x\right) \\ &= \sqrt{\frac{2}{\pi}}\left(\left[\frac{\sin(\omega x)}{\omega}\right]_0^1 - \left[\frac{\sin(\omega x)}{\omega}\right]_1^2\right) \\ &= \sqrt{\frac{2}{\pi}}\left(\frac{\sin\omega}{\omega} - \left(\frac{\sin 2\omega}{\omega} - \frac{\sin\omega}{\omega}\right)\right) \\ &= \sqrt{\frac{2}{\pi}}\left(\frac{2\sin\omega}{\omega} - \frac{\sin 2\omega}{\omega}\right). \end{aligned}$$

4.4 Exercise 4.

Find the Fourier transform of

$$f(x) = \begin{cases} e^{2ix}, & -1 < x < 1 \\ 0, & \text{otherwise.} \end{cases}.$$

We employ the definition of the Foruier transform

$$\mathscr{F}(f) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-i\omega x} \, \mathrm{d}x.$$

This becomes

$$\begin{aligned}
\mathscr{F}(f) &= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{2ix} e^{-i\omega x} \, \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{i(2-\omega)x} \, \mathrm{d}x \\
&= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{i(2-\omega)x}}{i(2-\omega)} \right]_{-1}^{1} \\
&= \frac{1}{\sqrt{2\pi}} \left(\frac{e^{i(2-\omega)} - e^{-i(2-\omega)}}{i(2-\omega)} \right) \\
&= \frac{1}{\sqrt{2\pi}} \frac{\cos(2-\omega) + i\sin(2-\omega) - \cos(\omega-2) + i\sin(\omega-2)}{i(2-\omega)} \\
&= \frac{1}{\sqrt{2\pi}} \frac{2i\sin(2-\omega)}{i(2-\omega)} \\
&= \sqrt{\frac{2}{\pi}} \frac{\sin(2-\omega)}{2-\omega}.
\end{aligned}$$

4.5 Exercise 5.

Use the formulae of the Fourier transform, Fourier cosine transform and Fourier sine transform for first and second derivatives from the Lecture to derive the corresponding formulae for third derivatives, i.e. for

$$\mathscr{F}_C\left(f'''\right), \mathscr{F}_S\left(f'''\right), \mathscr{F}\left(f'''\right).$$

We have that

$$\begin{aligned}
\mathscr{F}_C(f') &= \omega \mathscr{F}_S(f) - \sqrt{\frac{2}{\pi}} f(0) \\
\mathscr{F}_C(f'') &= -\omega^2 \mathscr{F}_C(f) - \sqrt{\frac{2}{\pi}} f'(0) \\
\mathscr{F}_S(f') &= -\omega \mathscr{F}_C(f) \\
\mathscr{F}_S(f'') &= -\omega^2 \mathscr{F}_S(f) + \sqrt{\frac{2}{\pi}} \omega f(0) \\
\mathscr{F}(f') &= i\omega \mathscr{F}(f) \\
\mathscr{F}(f'') &= -\omega^2 \mathscr{F}(f).
\end{aligned}$$

We will use that $f''' = (f'')'$ to get

$$\begin{aligned}
g' &= f''' \\
\mathscr{F}_C(f''') &= \mathscr{F}_C(g') \\
\mathscr{F}_C(g') &= \omega\mathscr{F}_S\left(g\right) - \sqrt{\frac{2}{\pi}}g(0) \\
\mathscr{F}_C\left(f'''\right) &= \omega\mathscr{F}_S(f'') - \sqrt{\frac{2}{\pi}}f''(0) \\
&= -\omega^3\mathscr{F}_S(f) + \sqrt{\frac{2}{\pi}}\omega^2 f(0) - \sqrt{\frac{2}{\pi}}f''(0).
\end{aligned}$$

The same procedure can now be followed for $\mathscr{F}_S(f''')$ to get:

$$\begin{aligned}
\mathscr{F}_S(f''') &= \mathscr{F}_S(g') \\
\mathscr{F}_S(g') &= -\omega\mathscr{F}_C(f) \\
\mathscr{F}_S(f''') &= -\omega\mathscr{F}_{\mathscr{C}}(f'') \\
&= \omega^3\mathscr{F}_C(f) + \sqrt{\frac{2}{\pi}}\omega f'(0).
\end{aligned}$$

For $\mathscr{F}(f''')$ we get

$$\begin{aligned}
\mathscr{F}(f''') &= \mathscr{F}(g') \\
\mathscr{F}(g') &= i\omega\mathscr{F}(f'') \\
&= i\omega\left(-\omega^2\mathscr{F}(f)\right) \\
&= -i\omega^3\mathscr{F}(f).
\end{aligned}$$

4.6 Exercise 6.

Let

$$\hat{f}_S(\omega) = \begin{cases} 1, & 0 < \omega < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Find $f(x)$.

We will use the definition of the inverse Fourier sine transform

$$f(x) = \mathscr{F}_S^{-1}(\hat{f}_S) = \sqrt{\frac{2}{\pi}}\int_0^\infty \hat{f}_S(\omega)\sin(\omega x)\,\mathrm{d}\omega.$$

In this case we have

$$\begin{aligned}
f(x) &= \sqrt{\frac{2}{\pi}}\int_0^1 \sin(\omega x)\,\mathrm{d}\omega \\
&= \sqrt{\frac{2}{\pi}}\left(-\frac{\cos x}{x} + \frac{1}{x}\right) \\
&= \sqrt{\frac{2}{\pi}}\left(\frac{1-\cos x}{x}\right).
\end{aligned}$$

4.7 Exercise 7.

Let

$$\hat{f}(\omega) = \begin{cases} 1, & -1 < \omega < 1 \\ 0, & \text{otherwise} \end{cases}.$$

Find $f(x)$.

We will now use the definition of the general Fourier transform

$$\begin{aligned} f(x) &= \mathscr{F}^{-1}(\hat{f}) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{i\omega x} \, \mathrm{d}\omega \\ &= \frac{1}{\sqrt{2\pi}} \int_{-1}^{1} e^{i\omega x} \, \mathrm{d}\omega \\ &= \frac{1}{\sqrt{2\pi}} \left[\frac{e^{i\omega x}}{ix} \right]_{-1}^{1} \\ &= \frac{1}{\sqrt{2\pi}} \frac{e^{ix} - e^{-ix}}{ix} \\ &= \frac{1}{\sqrt{2\pi}} \frac{\cos x + i \sin x - \cos(-x) - i \sin(-x)}{ix} \\ &= \frac{1}{\sqrt{2\pi}} \frac{2 \sin x}{x} \end{aligned}$$

.

4.8 Exercise 8.

Consider the ODE

$$f''(x) + f(x) = r(x)$$

where

$$r(x) = \begin{cases} 1, & 0 < x < 1 \\ -1, & 1 < x < 2 \\ 0, & x > 2 \end{cases}$$

and assume that we have the boundary condition $f'(0) = 0$. Apply the Fourier cosine transform to both sides of the ODE to derive the corresponding equation for $\hat{f}_C(\omega)$. Find $\hat{f}_C(\omega)$.

Hint: You may want to use the solution of Exercise 3.

Using linearity of Fourier transforms and the result form exercise 3 we get

$$\begin{aligned} \mathscr{F}_C(f'') + \mathscr{F}_C(f) &= \sqrt{\frac{2}{\pi}} \left(\frac{2 \sin \omega - \sin 2\omega}{\omega} \right) \\ -\omega^2 \hat{f}_C(\omega) + \hat{f}_C(\omega) &= \sqrt{\frac{2}{\pi}} \left(\frac{2 \sin \omega - \sin 2\omega}{\omega} \right) \\ \hat{f}_C(\omega) &= \sqrt{\frac{2}{\pi}} \frac{2 \sin \omega - \sin 2\omega}{\omega \left(1 - \omega^2\right)}. \end{aligned}$$

Lecture 5: Partial Differential Equations

26. September 2025

5.1 Exercise 1.

Write down the general expression for a linear first order PDE for $u(x, y, t)$.

From the lecture we see that the general form is

$$a_0(x,y,t) + a_1(x,y,t)u(x,y,t) + a_2(x,y,t)\frac{\partial u(x,y,t)}{\partial x} + a_3(x,y,t)\frac{\partial u(x,y,t)}{\partial y} + a_4(x,y,t)\frac{\partial u(x,y,t)}{\partial t} = 0.$$

5.2 Exercise 2.

Write down the general expression for a linear second order PDE for $u(x, y)$. You may assume that $u(x, y)$ is twice continuously differentiable.

From the lecture the general form is

$$\begin{aligned} 0 = a_0(x,y) + a_1(x,y)u(x,y) + a_2(x,y)\frac{\partial u(x,y)}{\partial x} + a_3(x,y)\frac{\partial u(x,y)}{\partial y} \\ + a_4(x,y)\frac{\partial^2 u(x,y)}{\partial x^2} + a_5(x,y)\frac{\partial^2 u(x,y)}{\partial y^2} + a_6(x,y)\frac{\partial u(x,y)}{\partial x \partial y} \end{aligned}$$

5.3 Exercise 3.

Consider the general homogeneous first order PDE for $u(x, t)$ with solutions $u_1(x, t)$ and $u_2(x, t)$. Show that $u(x, t) = c_1 u_1(x, t) + c_2 u_2(x, t)$ is also a solution for arbitrary constants c_1 and c_2.

We have given the following two solutions

$$\begin{aligned} a_1(x,t)u_1(x,t) + a_2(x,t)\frac{\partial u_1(x,t)}{\partial x} + a_3(x,t)\frac{\partial u_1(x,t)}{\partial t} = 0 \\ a_1(x,t)u_2(x,t) + a_2(x,t)\frac{\partial u_2(x,t)}{\partial x} + a_3(x,t)\frac{\partial u_2(x,t)}{\partial t} = 0. \end{aligned}$$

Now if we try to input $u(x, t) = c_1 u_1(x, t) + c_2 u_2(x, t)$ as our solution we get

$$\begin{aligned} 0 &= a_1 (c_1 u_1 + c_2 u_2) + a_2 \frac{\partial (c_1 u_1 + c_2 u_2)}{\partial x} + a_3 \frac{\partial (c_1 u_1 + c_2 u_2)}{\partial t} \\ &= c_1 \left(a_1 u_1 + a_2 \frac{\partial u_1}{\partial x} + a_3 \frac{\partial u_1}{\partial t} \right) + c_2 \left(a_1 u_2 + a_2 \frac{\partial u_2}{\partial x} + a_3 \frac{\partial u_2}{\partial t} \right) \\ &= 0 \end{aligned}$$

$\implies c_1 u_1 + c_2 u_2$ is a solution..

5.4 Exercise 4.

Find the solution of the one-dimensional wave equation

$$u_{tt}(x,t) = u_{xx}(x,t), \quad 0 \le x \le 1$$

subject to the boundary conditions

$$u(0,t) = u(1,t) = 0, \quad \text{for all } t$$

and the initial condition

$$u(x,0) = k_0 \sin(3\pi x), \quad u_t(x,0) = 0, \quad 0 \le x \le 1,$$

where k_0 is a constant. You may start your calculations directly from the general solution given in the Lecture.

The general solution of the wave equation is

$$u(x,t) = \sum_{n=1}^{\infty} \left(B_n \cos(\lambda_n t) + B_n^* \sin(\lambda_n t)\right) \sin(n\pi x), \qquad \lambda_n = n\pi.$$

We use the initial elongation to get

$$u(x,0) = \sum_{n=1}^{\infty} (B_n) \sin(n\pi x) = k_0 \sin(3\pi x) \implies B_3 = k_0.$$

All other coefficients B_n are zero. The initial velocity now gives

$$u_t(x,0) = B_n^* \sin(n\pi x) = 0 \implies B_n^* = 0.$$

Therefore the unique solution becomes

$$u(x,t) = k_0 \cos(3\pi t) \sin(3\pi x).$$

5.5 Exercise 5.

Find the solution of the one-dimensional wave equation

$$u_{tt}(x,t) = u_{xx}(x,t), 0 \le x \le 1$$

subject to the boundary conditions

$$u(0,t) = u(1,t) = 0, \quad \text{for all } t$$

and the initial condition

$$u(x,0) = k_0 \sin(3\pi x), \quad u_t(x,0) = 3\pi k_0 \sin(3\pi x), 0 \le x \le 1,$$

where k_0 is a constant. You may start your calculation directly from the general solution given in the Lecture.

The general solution of the wave equation is

$$u(x,t) = \sum_{n=1}^{\infty} \left(B_n \cos(\lambda_n t) + B_n^* \sin(\lambda_n t)\right) \sin(n\pi x), \qquad \lambda_n = n\pi.$$

The initial elongation gives us

$$u(x,0) = \sum_{n=1}^{\infty} B_n \sin(n\pi x) = k_0 \sin(3\pi x) \implies B_3 = k_0.$$

The initial velocity gives

$$u_t(x,0) = \sum_{n=1}^{\infty} B_n^* \lambda_n \sin(n\pi x) = 3\pi k_0 \sin(3\pi x) \implies B_3^* = k_0.$$

Which corresponds to the solution

$$u(x,t) = (k_0 \cos(3\pi t) + k_0 \sin(3\pi t))\sin(3\pi x).$$

Lecture 6: Heat equation

3. Oktober 2025

6.1 Exercise 1.

Consider the one-dimensional heat equation

$$u_t(x,t) = u_{xx}(x,t), 0 \le x \le 1$$

with boundary conditions

$$u(0,t) = u(1,t) = 0$$

for all t. Suppose you have the initial temperature profiles

$$u(x,0) = \sin(\pi x)$$

and the approximation (reflecting the situation where you do not know all details of the initial temperature profile)

$$u_N(x,0) = \sin(\pi x) + \sin(N\pi x)$$

where N is an integer number. Let $u(x,t)$ and $u_N(x,t)$ be the corresponding solutions to the above heat equation. Find the integrated squared error between these solutions (integration with respect to $0 \le x \le 1$) as a function of time.

From the initial condition we have

$$u(x,0) = \sin(\pi x) \implies B_1 = 1.$$

Meaning the solution becomes

$$u(x,t) = e^{-\lambda_1^2 t}\sin(\pi x).$$

And

$$u_N(x,0) = \sin(\pi x) + \sin(N\pi x) \implies B_1 = B_N = 1.$$

Meaning the solution is

$$u_N(x,t) = e^{-\lambda_1^2 t}\sin(\pi x) + e^{-\lambda_N^2 t}\sin(N\pi x).$$

The integrated squared error then is

$$\begin{aligned} E_N(t) &= \int_0^1 (u(x,t) - u_N(x,t))^2 \, \mathrm{d}x \\ &= e^{-2\lambda_N^2 t}\int_0^1 \sin^2(N\pi x)\, \mathrm{d}x \\ &= \frac{1}{2}e^{-2\lambda_N^2 t}. \end{aligned}$$

6.2 Exercise 2.

Consider the one-dimensional heat equation

$$u_t(x,t) = u_{xx}(x,t), 0 \leq x \leq L$$

with boundary conditions

$$u_x(0,t) = u_x(L,t) = 0$$

for all t and initial condition

$$u_t(x,0) = h(x), 0 \leq x \leq L.$$

Find the general solution for this situation.

The general solution that fulfils the boundary conditions are

$$u(x,t) = B_0 + \sum_{n=1}^{\infty} B_n e^{-\lambda_n^2 t} \cos\left(\frac{n\pi x}{L}\right).$$

The derivative at $t = 0$ is given as $h(x)$. This can also be found from the above as

$$u_t(x,0) = \sum_{n=1}^{\infty} \left(-\lambda_n^2 B_n\right) \cos\left(\frac{n\pi x}{L}\right).$$

From the formula for the Fourier coefficients we get

$$B_n = -\frac{2}{\lambda_n^2 L} \int_+^L h(x) \cos\left(\frac{n\pi x}{L}\right) \mathrm{d}x.$$

Which means the general solution that fulfills the boundary conditions are

$$u(x,t) = B_0 + \sum_{n=1}^{\infty} \left(-\frac{2}{\lambda_n^2 L} \int_0^L h(x) \cos\left(\frac{n\pi x}{L}\right) \mathrm{d}x\right) e^{-\lambda_n^2 t} \cos\left(\frac{n\pi x}{L}\right).$$

With arbitrary B_0.

6.3 Exercise 3.

Consider the PDE

$$u_{xx}(x,t) = u_{tt}(x,t), 0 \leq x \leq 1.$$

(a) Find the general solution that fulfils the conditions

$$u_x(0,t) = u_x(1,t) = 0$$

for all t.

From separation of variables we get

$$u(x,t) = F(x)G(t) \implies \frac{F''(x)}{F(x)} = \frac{G''(t)}{G(t)} = k.$$

The boundary conditions gives

$$\begin{aligned}
u_x(0,t) &= F'(0)G(t) = 0 \implies F'(0) = 0 \\
u_x(1,t) &= F'(1)G(t) = 0 \implies F'(1) = 0.
\end{aligned}$$

Now for $k < 0$ we get

$$\begin{aligned}
F(x) &= A\cos(px) + B\sin(px), \qquad p = \sqrt{-k} \\
F'(x) &= -Ap\sin(px) + Bp\cos(px) \\
F'(0) &= B_p = 0 \implies B = 0 \\
F'(1) &= -Ap\sin p = 0 \implies p = p_n = n\pi, \qquad n = 1, 2, \ldots \\
&\implies F_n(x) = \cos(n\pi x) \text{ for } A = 1.
\end{aligned}$$

For $k = 0$ we get

$$\begin{aligned}
F(x) &= B + Ax \\
F'(0) &= 0 \implies A = 0 \implies F'(1) = 0 \\
&\implies F(x) = 1 \text{ for } B = 1.
\end{aligned}$$

For $k > 0$ we get

$$\begin{aligned}
F(x) &= Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x} \\
F'(x) &= A\sqrt{k}e^{\sqrt{k}x} - B\sqrt{k}e^{-\sqrt{k}x} \\
F'(0) &= A\sqrt{k} - B\sqrt{k} = 0 \implies A = B \\
F'(1) &= A\sqrt{k}\left(e^{\sqrt{k}} - e^{-\sqrt{k}}\right) = 0 \implies A = B = 0 \\
&\implies F(x) = 0
\end{aligned}$$

.

Hence only the cases for $k = 0$ and $k < 0$ are interesting.

The spatial solutions become

$$\begin{aligned}
F_0(x) &= 1 && k = 0 \\
F_n(x) &= \cos(n\pi x) && k < 0, k = -p_n^2.
\end{aligned}$$

The temporal solutions become

$$\begin{aligned}
G_0(t) &= B_0 + B_0^* t && k = 0 \\
G_n(t) &= B_n\cos(n\pi t) + B_n^*\sin(n\pi t) && k < 0, k = -p_n^2.
\end{aligned}$$

And the fundamental solutions become

$$\begin{aligned}
u_0(x,t) &= G_0(t)F_0(x) = B_0 + B_0^* t && k = 0 \\
u_n(x,t) &= G_n(t)F_n(x) = (B_n\cos(n\pi t) + B_n^*\sin(n\pi t))\cos(n\pi x) && k < 0, k = -p_n^2.
\end{aligned}$$

Which corresponds to the general solution

$$\begin{aligned}u(x,t) &= u_0(x,t) + \sum_{n=1}^{\infty} u_n(x,t) \\ &= B_0 + B_0^* t + \sum_{n=1}^{\infty} \big(B_n \cos(n\pi t) + B_n^* \sin(n\pi t)\big)\cos(n\pi x)\end{aligned}$$

(b) Find the general solution that fulfils the conditions

$$u_x(0,t) = u_x(1,t) = 0$$

for all t and

$$u(x,0) = 1 + \cos(3\pi x)$$

for $0 \le x \le 1$.

Here we have the additional condition compared to above of

$$u(x,0) = B_0 + \sum_{n=1}^{\infty} B_n \cos(n\pi x) = 1 + \cos(3\pi x) \implies B_0 = B_3 = 1.$$

Which gives the general solution

$$u(x,t) = 1 + B_0^* t + \cos(3\pi t)\cos(3\pi x) + \sum_{n=1}^{\infty} B_n^* \sin(n\pi t)\cos(n\pi x).$$

6.4 Exercise 4.

Consider the one-dimensional heat equation

$$u_t(x,t) = u_{xx}(x,t), 0 \le x \le L$$

with boundary conditions

$$u(0,t) = u(L,t) = 0.$$

(a) Calculate the average temperature

$$\frac{1}{L}\int_0^L u(x,t)\,\mathrm{d}x$$

for the initial condition

$$u(x,0) = \sin\left(\frac{\pi x}{L}\right).$$

The solution is

$$u(x,0) = \sin\left(\frac{\pi x}{L}\right) \implies u(x,t) = e^{-\lambda_1^2 t}\sin\left(\frac{\pi x}{L}\right).$$

And thus, the average temperature is

$$\frac{1}{L}\int_0^L u(x,t)\,\mathrm{d}x = \frac{2}{\pi}e^{-\lambda_1^2 t}.$$

(b) Calculate the average temperature for the initial condition

$$u(x,0) = \sin\left(\frac{\pi x}{L}\right) + \sin\left(\frac{2\pi x}{L}\right).$$

Here the solution is

$$\begin{aligned} u(x,o) &= \sin\left(\frac{\pi x}{L}\right) + \sin\left(\frac{2\pi x}{L}\right) \\ \implies u(x,t) &= e^{-\lambda_1^2 t} \sin\left(\frac{\pi x}{L}\right) + e^{-\lambda_2^2 t} \sin\left(\frac{2\pi x}{L}\right). \end{aligned}$$

Which gives the average temperature

$$\frac{1}{L}\int_0^L u(x,t)\,\mathrm{d}x = \frac{2}{\pi} e^{-\lambda_1^2 t}.$$

Lecture 7: The heat equation continued

24. Oktober 2025

Problem 7.1

Consider the one-dimensional heat equation

$$u_t(x,t) = c^2 u_{xx}(x,t), 0 \le x \le L$$

subject to the boundary conditions

$$u(0,t) = u(L,t) = 0$$

for all t.

Let T_0 be a constant and

$$u(x,0) = \begin{cases} T_0, & 0 < x < L \\ 0, & x = 0 \\ 0, & x = L. \end{cases}$$

This set-up can be used to describe the situation where a very thin, latterally insulated bar of constant temperature T_0 is, at time $t = 0$, thrown into a heat bath of zero temperature. Find the solution $u(x,t)$.

Derivation

From the lecture, the general solution to the one-dimensional heat equation with the given boundary conditions, is

$$u(x,t) = \sum_{n=1}^{\infty} B_n e^{-\lambda_n^2 t} \sin\left(\frac{n\pi x}{L}\right).$$

Where B_n is:

$$\begin{aligned} B_n &= \frac{2}{L}\int_0^L u(x,0)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x \\ &= \frac{2}{L}\int_0^L T_0 \sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x \\ &= \frac{2T_0}{L}\int_0^L \sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x \\ &= -\frac{2T_0}{L}\left(\cos(n\pi) - 1\right). \end{aligned}$$

Hence the solution becomes:

$$\boxed{u(x,t) = \sum_{n=1}^{\infty} e^{-\lambda_n^2 t} \frac{2T_0}{L}\left(1 - \cos(n\pi)\right)\sin\left(\frac{n\pi x}{L}\right)}$$

RESULT

Problem 7.2

Consider the one-dimensional heat equation

$$u_t(x,t) = c^2 u_{xx}(x,t), 0 \le x \le L$$

subject to the boundary conditions

$$u(0,t) = T_1, u(L,t) = T_2$$

for all t.

Let T_0 be a constant and

$$u(x,0) = \begin{cases} \frac{T_2 - T_1}{L}x + T_0, & 0 < x < L \\ T_1, & X = 0 \\ T_2, & x = L. \end{cases}$$

Find the solution $u(x,t)$.

Problem 7.3

Consider the steady heat flow on a thin plate in the xy-plane, $0 \le x \le 1, 0 \le y \le 1$, insulated from above/below, governed by

$$u_{xx}(x,y) + u_{yy}(x,y) = 0.$$

Find the solution that corresponds to the boundary conditions

$$\begin{aligned} &u(x,0) = 0, && u(x,1) = 0, && 0 \le x \le 1 \\ &u(0,y) = 0, && u(1,y) = 0, && 0 \le y \le 1. \end{aligned}$$

Problem 7.4

Consider the steady heat flow on a thin plate in the xy-plane, $0 \le x \le 3, 0 \le y \le 2$, insulated from above/below, governed by

$$u_{xx}(x,y) + u_{yy}(x,y) = 0.$$

Find the solution that corresponds to the boundary conditions

$$\begin{aligned} &u(x,0) = 0, && u(x,2) = 2\sin\frac{4\pi x}{3}, && 0 \le x \le 3 \\ &u(0,y) = 0, && u(3,y) = 0, && 0 \le y \le 2. \end{aligned}$$

Problem 7.5

Consider the steady heat flow on a thin plate in the xy-plane, $0 \le x \le 1, 0 \le y \le 1$, insulated from above/below, governed by

$$u_{xx}(x,y) + u_{yy}(x,y) = 0.$$

Find the solution that corresponds to the boundary conditions

$$\begin{aligned} &u(x,0) = 0, \qquad && u_y(x,1) = 0, \qquad && 0 \le x \le 1 \\ &u(0,y) = 0, && u(1,y) = 0, && 0 \le y \le 1. \end{aligned}$$

Lecture 8: Vibrating rectangular membranes and Double Fourier Sine Series 31. Oktober 2025

Problem 8.1

Consider the function

$$f(x,y) = 3\sin\left(\frac{4\pi y}{3}\right)\sin\left(\frac{3\pi x}{2}\right) + 2\sin\left(\frac{4\pi y}{3}\right)\sin\left(\frac{3\pi x}{6}\right), 0 \le x \le 2, 0 \le y \le 3.$$

Find the Fourier coefficients of the double Fourier sine series of $f(x,y)$.

Hint: Coefficients of double Fourier sine series are unique.

Problem 8.2

Consider the deflection $u(x,t)$ of a vibrating elastic membrane in the xy-plane, $0 \le x \le 1, 0 \le y \le 1$, fixed along its boundaries and governed by

$$u_{tt}(x,y,t) = c^2\left(u_{xx}(x,y,t) + u_{yy}(x,y,t)\right).$$

(a) Find the solution that corresponds to the initial conditions

$$\begin{aligned} u(x,y,0) &= \sin(6\pi x)\sin(2\pi y) \\ u_t(x,y,0) &= 0. \end{aligned}$$

(b) Find the solution that corresponds to the initial conditions

$$\begin{aligned} u(x,y,0) &= 0 \\ u_t(x,y,0) &= \sin(6\pi x)\sin(2\pi y). \end{aligned}$$

(c) Find the solution that corresponds to the initial conditions

$$\begin{aligned} u(x,y,0) &= \sin(6\pi x)\sin(2\pi y) \\ u_t(x,y,0) &= \sin(6\pi x)\sin(2\pi y). \end{aligned}$$

Hint: Coefficients of double Fourier dine series are unique.

Problem 8.3

Consider the two-dimensional heat equation

$$u_t(x, y, t) = c^2(u_{xx}(x, y, t) + u_{yy}(x, y, t)).$$

(a) Assume $u(x, y, t) = F(x, y)G(t)$ and derive a PDE for $F(x, y)$ and an ODE for $G(t)$.

(b) Assume $F(x, y) = H(x)Q(y)$ to finally derive three ODEs for $H(x)$, $Q(y)$ and $G(t)$.

Problem 8.4

Consider the PDE

$$u_t(x, y, z, t) = u_x(x, y, z, t) + u_y(x, y, z, t) + u_z(x, y, z, t).$$

Assume $u(x, y, z, t) = H(x)Q(y)P(z)G(t)$ and derive 4 ODEs for $H(x), Q(y), P(z)$, and $G(t)$.

Lecture 9: D'Alembert's solution of the one-dimensional wave equation 7. November 2025

Problem 9.1

Calculate the second derivatives of

$$u(x, t) = \Phi(x + ct) + \Psi(x - ct)$$

with respect to t and x and show that any function of this type is a solution to the one-dimensional wave equation

$$u_{tt}(x, t) = c^2 u_{xx}(x, t)$$

for arbitrary twice differentiable functions Φ and Ψ.

Problem 9.2

Consider the PDE

$$u_{xt}(x, t) = 0$$

and the change of variables

$$v = v(x, t) = x + 2t, \qquad w = w(x, t) = x - 2t$$
$$x = x(v, w) = \frac{1}{2}(v + w), \qquad t = t(v, w) = \frac{1}{4}(v - w).$$

Derive the corresponding PDE for

$$\overline{u}(v, w) = u(x(v, w), t(v, w)).$$

You may assume that $\overline{u}(v, w)$ is twice continuously differentiable, i.e. $\overline{u}_{vw}(v, w) = \overline{u}_{wv}(v, w)$.

Problem 9.3

Calculate d'Alembert's solution corresponding to the initial conditions

$$u(x,0) = k_0 \sin(3\pi x)\,, u_t(x,0) = 3\pi k_0 \sin(3\pi x)\,, -\infty < x < \infty,$$

where k_0 is a constant.

Compare your result to that of Exercice 5, Lecture 5.

Hint: $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta, \cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$.